



## Cloud Computing

**Name: Fatima Jaleel**

**Reg #: FA24B1-SE-037**

**Section: VB**

**Course Code:SE-665**

**GitHub username: fatimajaleel**

### LAB # 01

---

#### Lab Title:

GitHub Identity, Student Portal Registration, and Ubuntu Server Installation

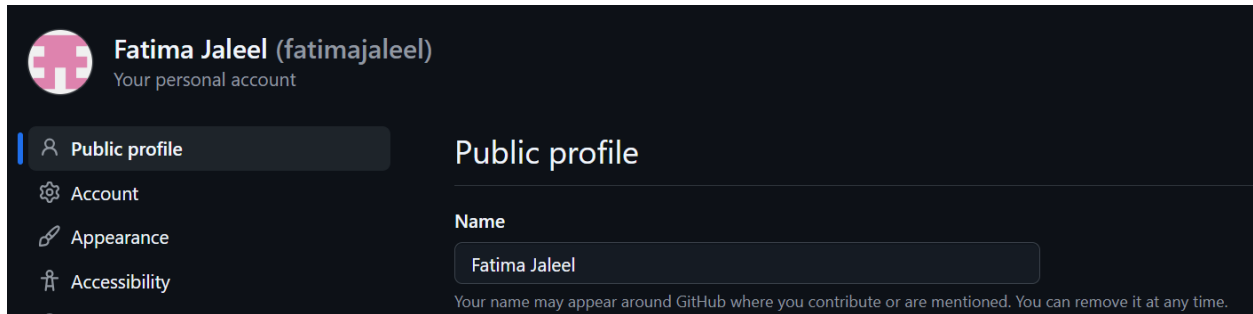
#### Task 1: Create and Configure Your GitHub Account

#### Screenshot 1:



GitHub profile showing my profile name and username

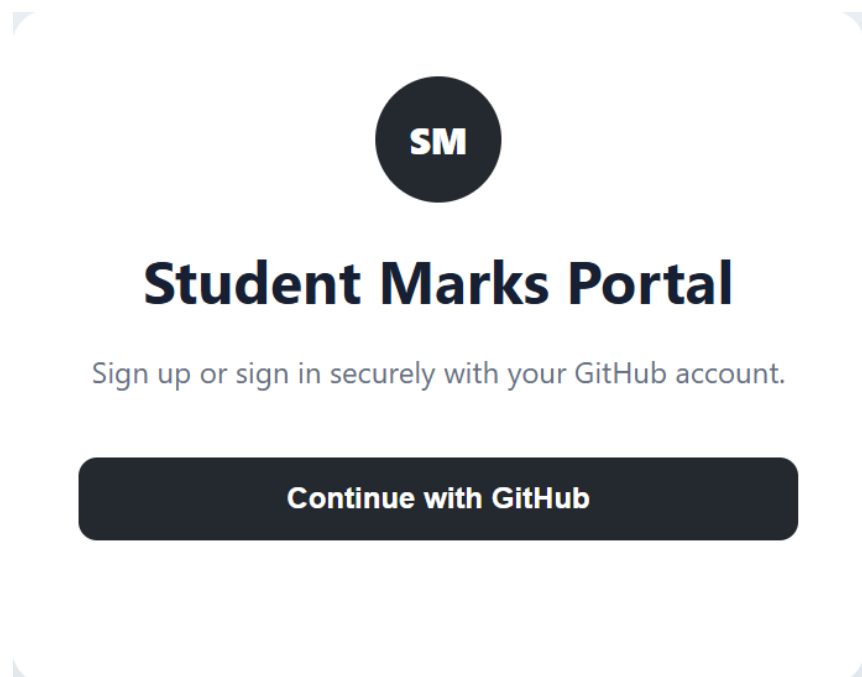
## Screenshot 2:



GitHub Public Profile showing my complete actual name

## Task 2: Register in the Student Marks Portal

### Screenshot 1:



Student Marks Portal login page

Screenshot 2:

STUDENT MARKS PORTAL

Course enrollments

Sign out



Fatima Jaleel

2024-BSE-37

+ Enroll in another course

ENROLLMENT STATUS

Your courses

Cloud Computing (SE-665) Section B

Waiting for administrator approval

Pending approval

Screenshot 3:

+ Enroll in another course

APPROVED COURSES

Course Marks

Published

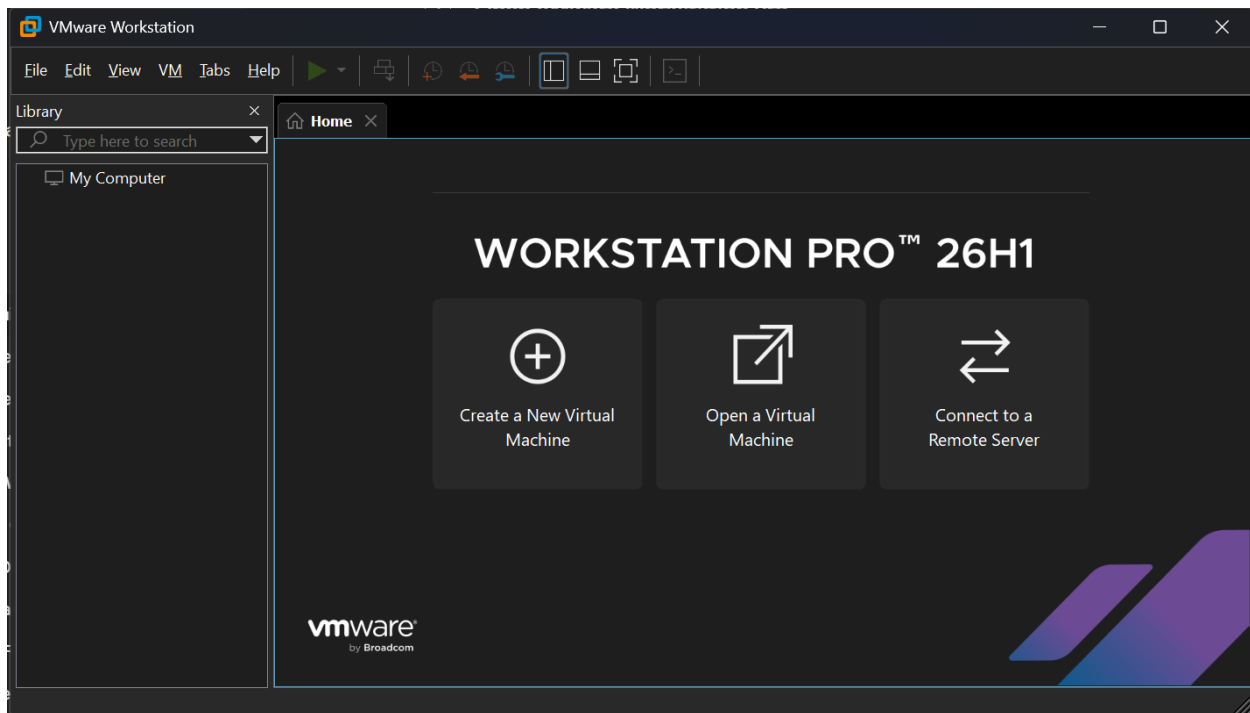
Cloud Computing (SE-665) Section B

Class Participation

| ASSESSMENT          | OBTAINED | TOTAL | PERCENTAGE |
|---------------------|----------|-------|------------|
| Class Participation | 1        | 10    | 10.0%      |
| Total               | 1        | 10    | 10.0%      |

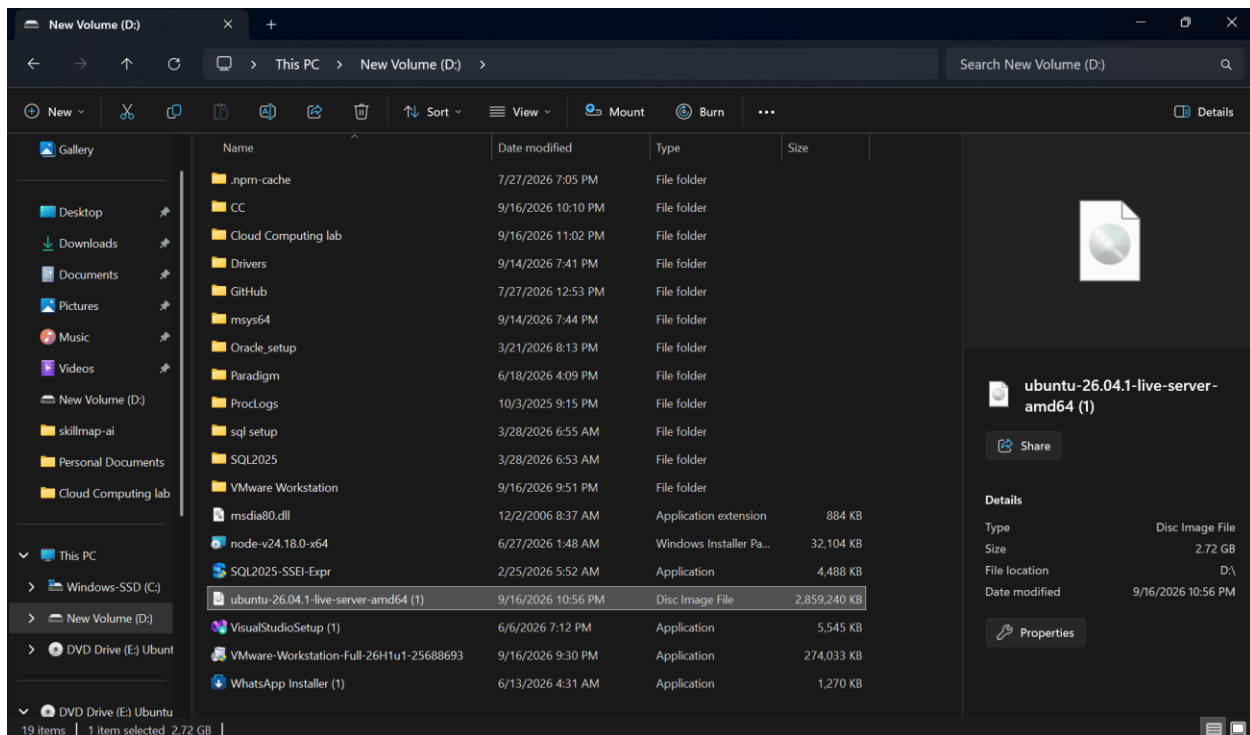
Task 3: Download VMware Workstation Pro and Ubuntu Server

Screenshot 1:



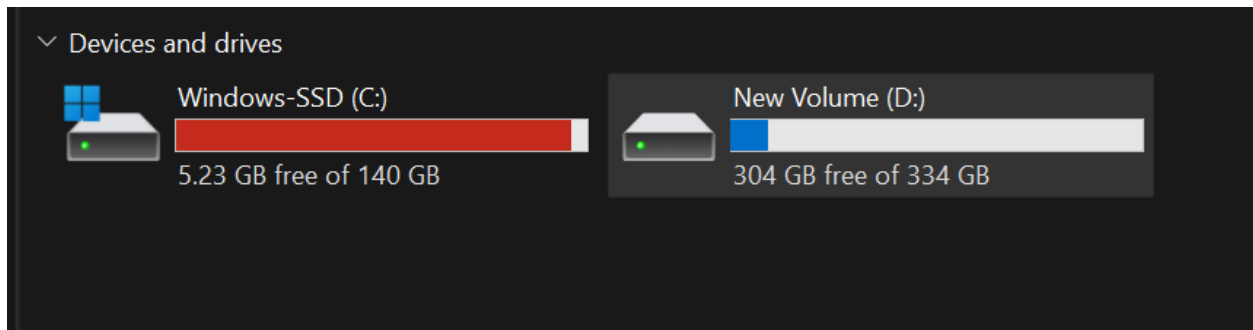
VMware Workstation Pro successfully running.

## Screenshot 2:



Downloaded Ubuntu Server ISO filename.

### Screenshot 3:



Available storage showing at least 20 GB free space

## Task 4: Create the Ubuntu Server Virtual Machine

### 1. New VM Wizard:



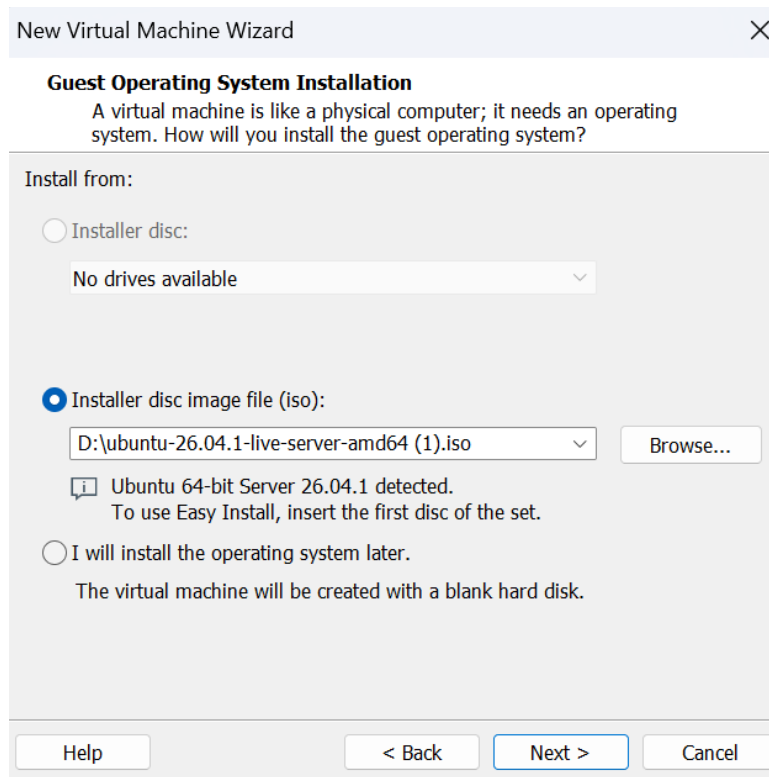
VMware New Virtual Machine Wizard.

### 2. Typical configuration:



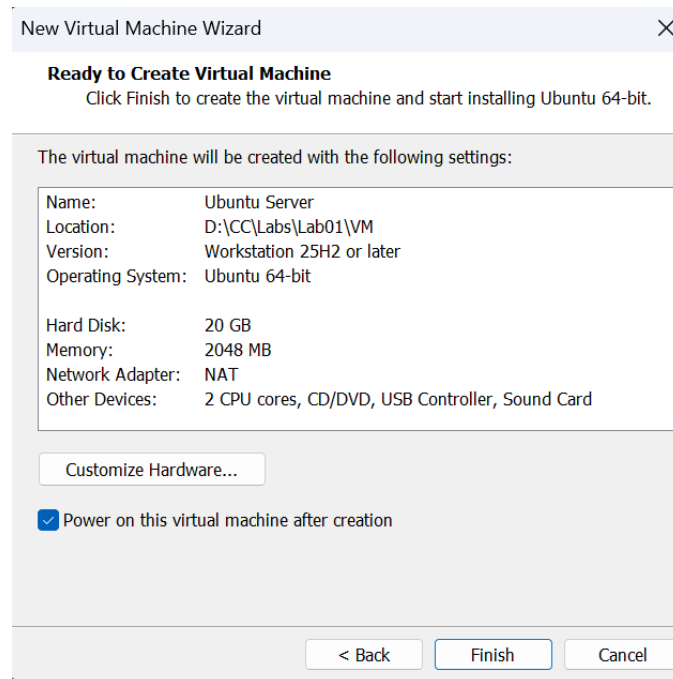
Typical virtual machine configuration selected.

### 3. Ubuntu ISO selected:



Ubuntu Server ISO selected in VMware.

## 4. Configuration Summary:



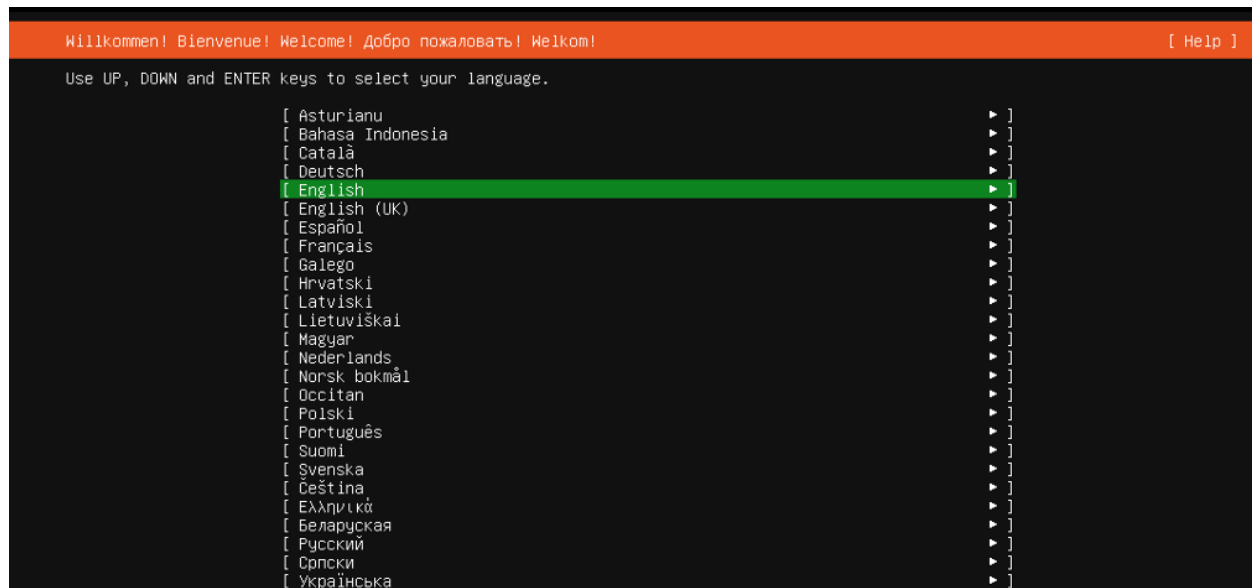
Final Ubuntu Server virtual machine configuration.

## 5. Ubuntu boot screen:

```
[ OK ] Mounted snap-hwctl-123.mount - Mount unit for hwctl, revision 123.
[ OK ] Mounted snap-snapd-27710.mount - Mount unit for snapd, revision 27710.
[ OK ] Reached target snapd.mounts.target - Mounted snaps.
[ OK ] Reached target local-fs.target - Local File Systems.
[ OK ] Listening on systemd-sysext.socket. System Extension Image Management.
Starting apparmor.service - Load AppArmor profiles...
Starting console-setup.service - Set console font and keymap...
Starting finalrd.service - Create .time dir for shutdown pivot root...
Starting kdump-tools.service - Kernel crash dump capture service...
Starting netplan-config.service - Netplan Backend Configuration...
Starting plymouth-read-write.service - Plymouth To Write Out Runtime Data...
Starting systemd-binfmt.service - Set Up Additional Binary Formats...
Starting systemd-tmpfiles-setup.service - Create System Files and Directories...
Starting ufw.service - Uncomplicated firewall...
[ OK ] Finished console-setup.service - Set console font and keymap.
[ OK ] Finished finalrd.service - Create .untime dir for shutdown pivot root.
[ OK ] Finished ufw.service - Uncomplicated firewall.
[ OK ] Finished plymouth-read-write.service - Plymouth To Write Out Runtime Data.
[ OK ] Finished systemd-tmpfiles-setup.service - Create System Files and Directories.
[ OK ] Finished kdump-tools.service - Kernel crash dump capture service.
[ OK ] Finished netplan-config.service - Netplan Backend Configuration.
[ OK ] Reached target network-pre.target - Preparation for Network.
[ OK ] Listening on systemd-networkd-resolve.socket. Network Management Resolve Hook Socket.
Starting systemd-networkd.service - Network Management...
```

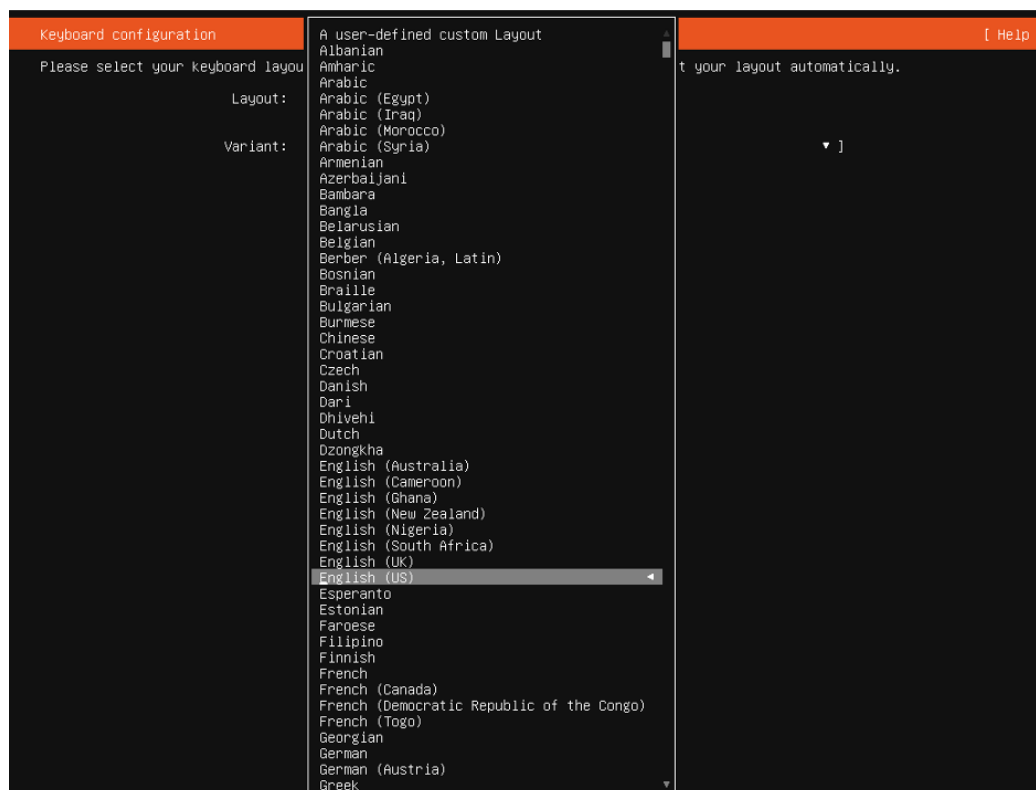
Ubuntu Server installer boot screen.

## Task 5: Choose the Installation Language:



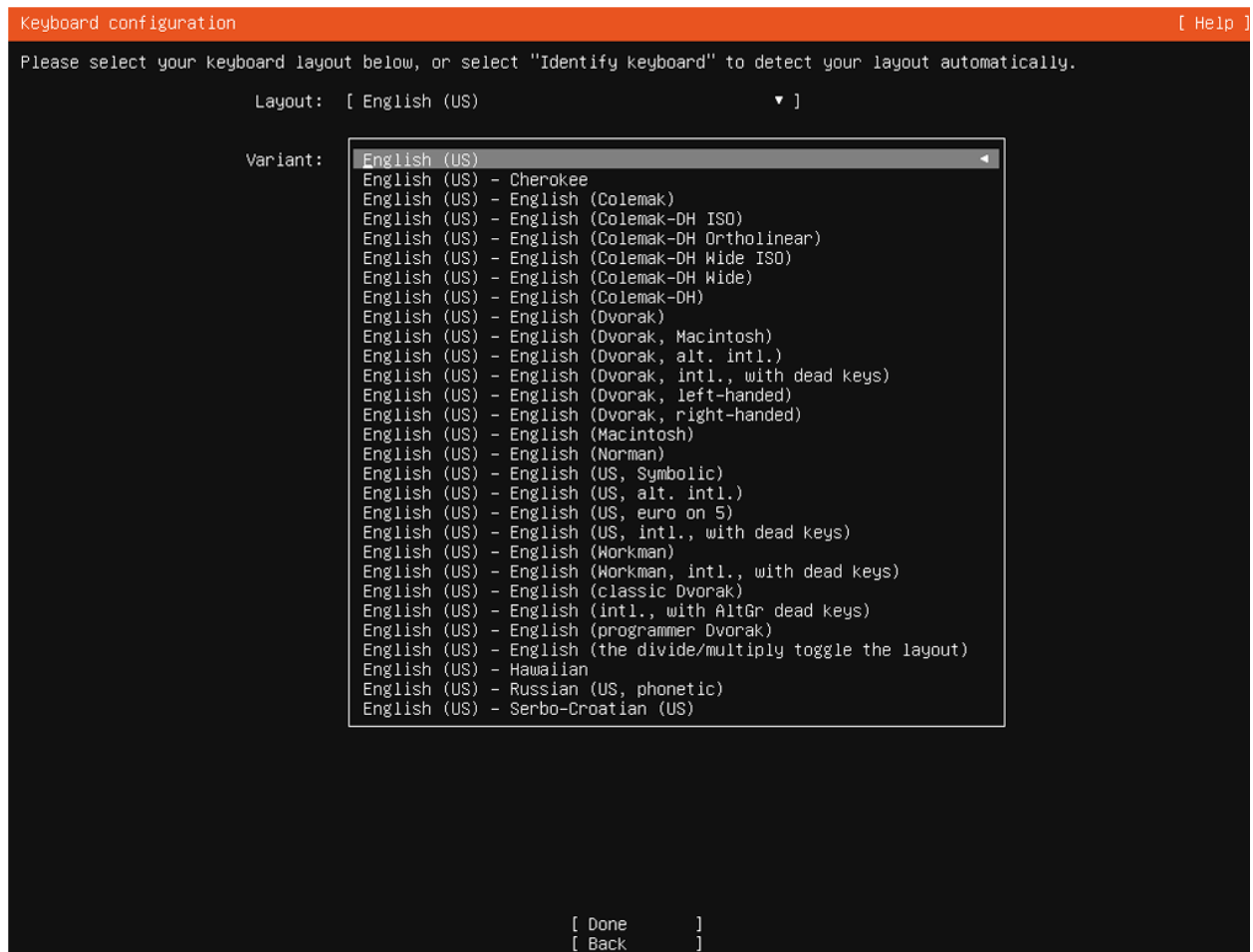
Selected Ubuntu Server installation language.

## Task 6: Configure the Keyboard Layout:



Selected keyboard layout.

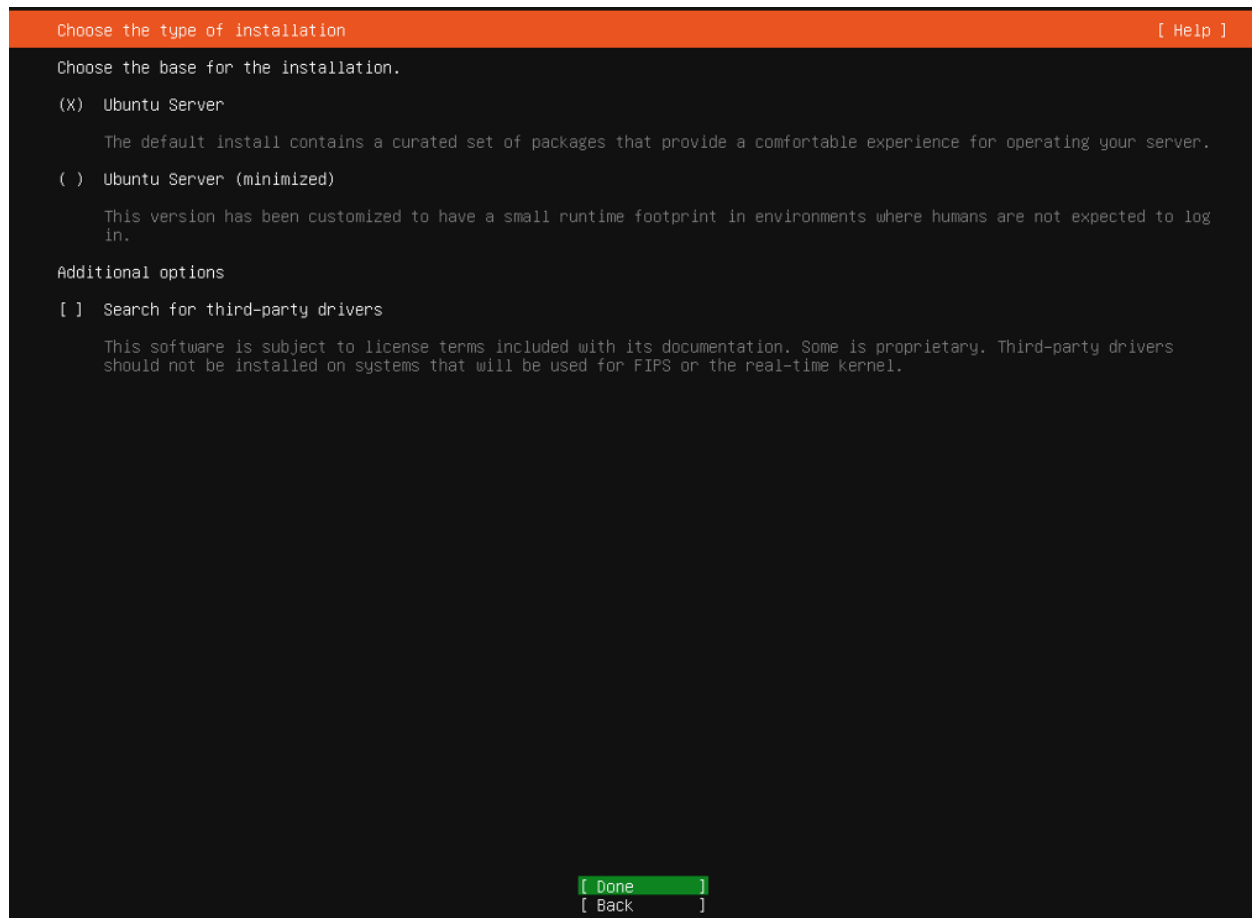
## Screenshot 2:



Selected keyboard variant.

## Task 7: Choose the Ubuntu Server Installation Type

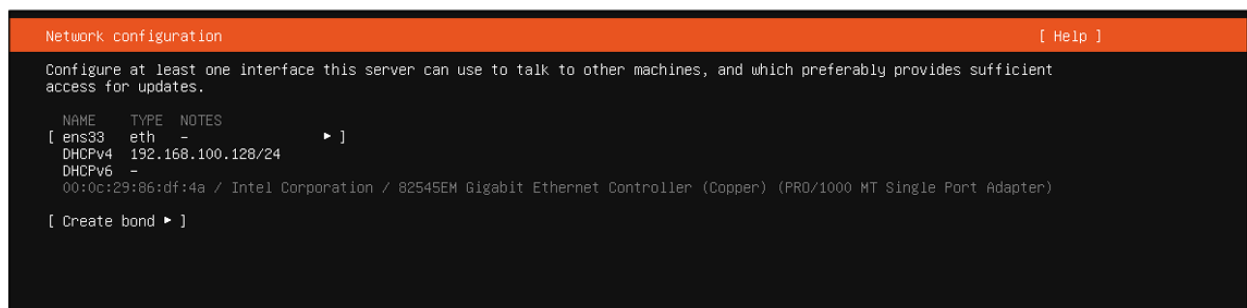
### Screenshot 1:



Normal Ubuntu Server installation option selected.

## Task 8: Configure the Network

### Screenshot 1:



Ubuntu Server detected network interface.

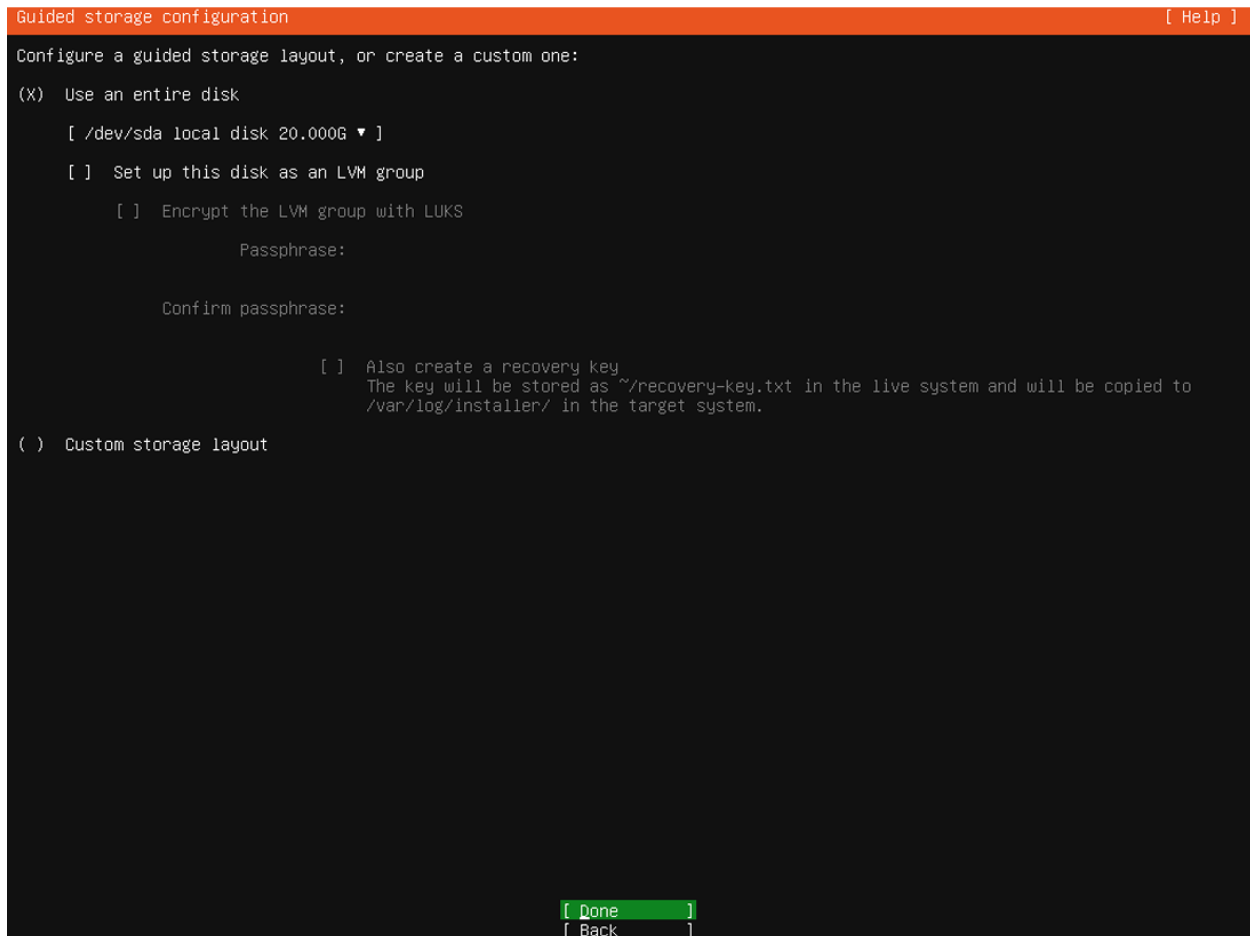
### Screenshot 2:

```
NAME      TYPE  NOTES
[ ens33   eth   -      ▶ ]
DHCPv4    192.168.100.128/24
DHCPv6    -
00:0c:29:86:df:4a / Intel Corporation / 82545EM Gigabit Ethernet Controller (Copper) (PRO/1000 MT Single Port Adapter)
```

Ubuntu Server network interface showing assigned IP address.

## Task 9: Configure Storage

### Screenshot 1:



Guided storage option selected.

### Screenshot 2:



Ubuntu Server storage configuration.

## Task 10: Select the Installation Disk

### Screenshot 1:

| DEVICE     | TYPE       | SIZE        |
|------------|------------|-------------|
| [ /dev/sda | local disk | 20.000G ▶ ] |

Selected VMware virtual disk.

### Screenshot 2:

## USED DEVICES

| DEVICE      | TYPE                                       | SIZE    |     |
|-------------|--------------------------------------------|---------|-----|
| [ /dev/sda  | local disk                                 | 20.000G | ▶ ] |
| partition 1 | new, BIOS grub spacer                      | 1.000M  | ▶   |
| partition 2 | new, to be formatted as ext4, mounted at / | 19.997G | ▶   |

Ubuntu virtual disk name and size.

## Task 11: Review the Partitions

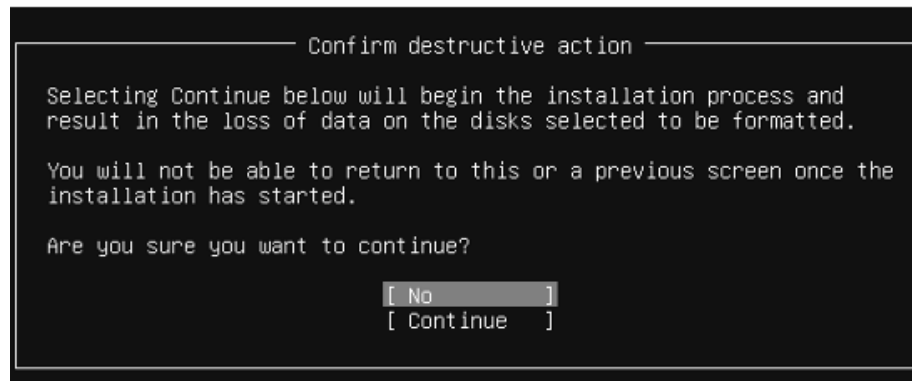
### Screenshot 1:



Proposed Ubuntu partition layout

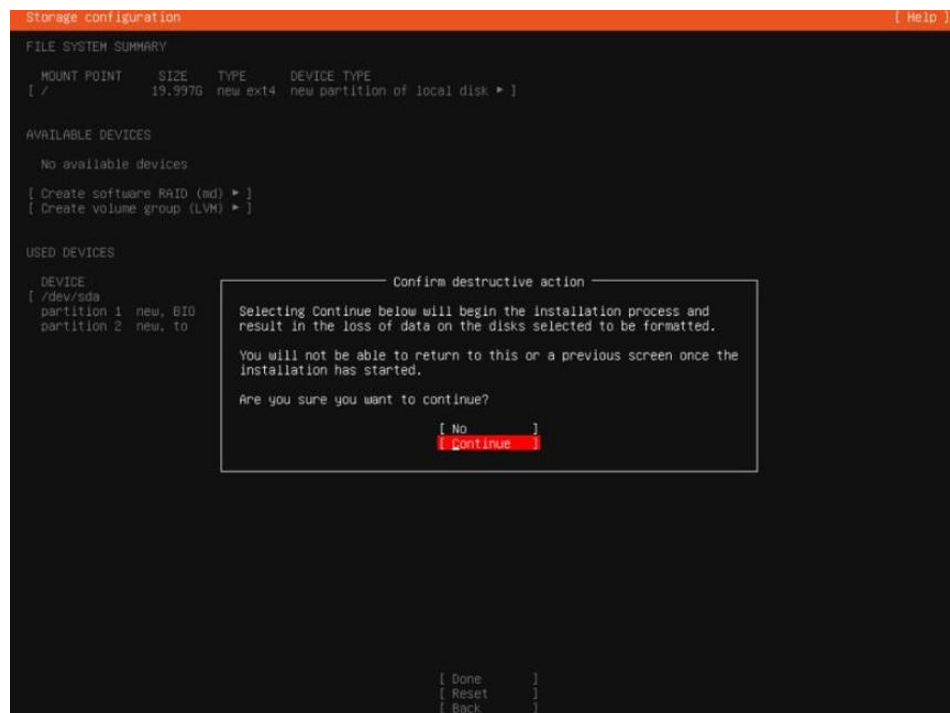
## Task 12: Confirm the Storage Changes

Screenshot 1:



Final storage warning.

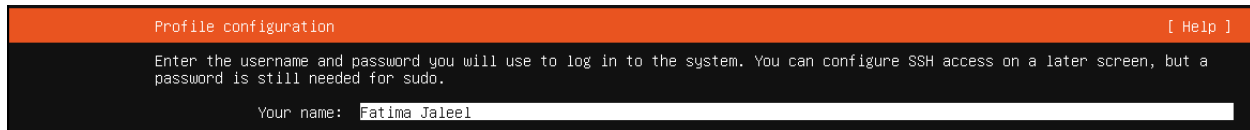
Screenshot 2:



Storage changes confirmation

## Task 13: Configure the Ubuntu Profile

Screenshot 1:



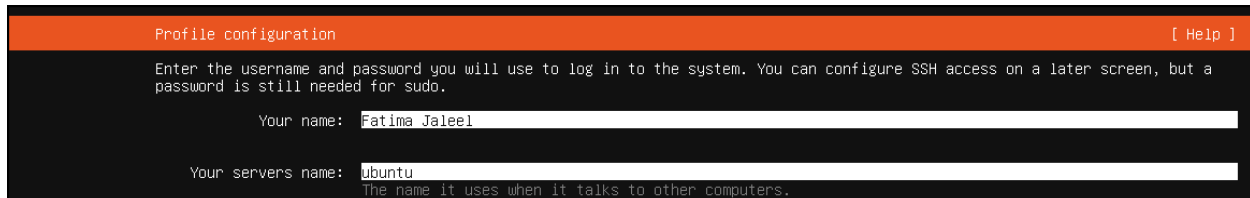
Profile configuration [ Help ]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Ubuntu profile showing complete actual name

### Screenshot 2:



Profile configuration [ Help ]

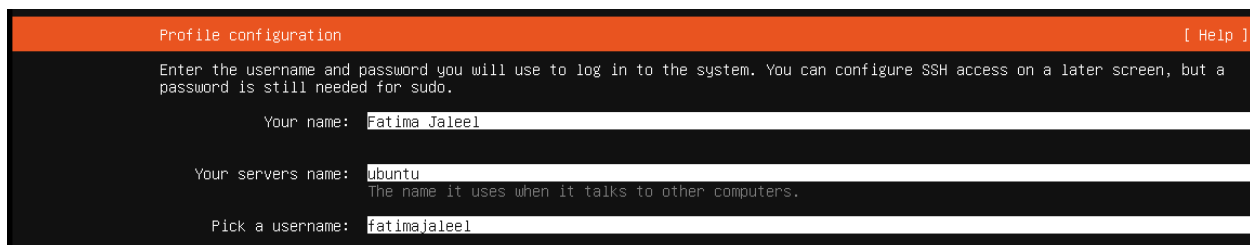
Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Your servers name:   
The name it uses when it talks to other computers.

Ubuntu server name set to ubuntu

### Screenshot 3:



Profile configuration [ Help ]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

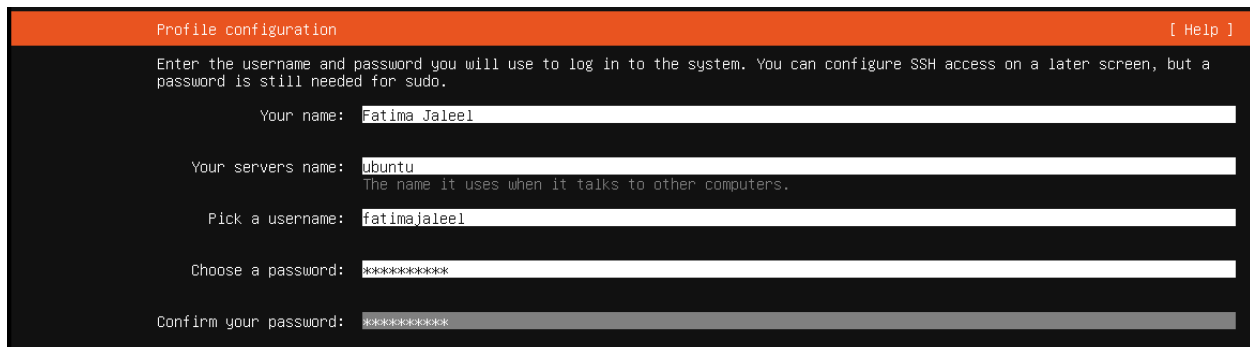
Your name:

Your servers name:   
The name it uses when it talks to other computers.

Pick a username:

Ubuntu username matching GitHub username

### Screenshot 4:



Profile configuration [ Help ]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Your servers name:   
The name it uses when it talks to other computers.

Pick a username:

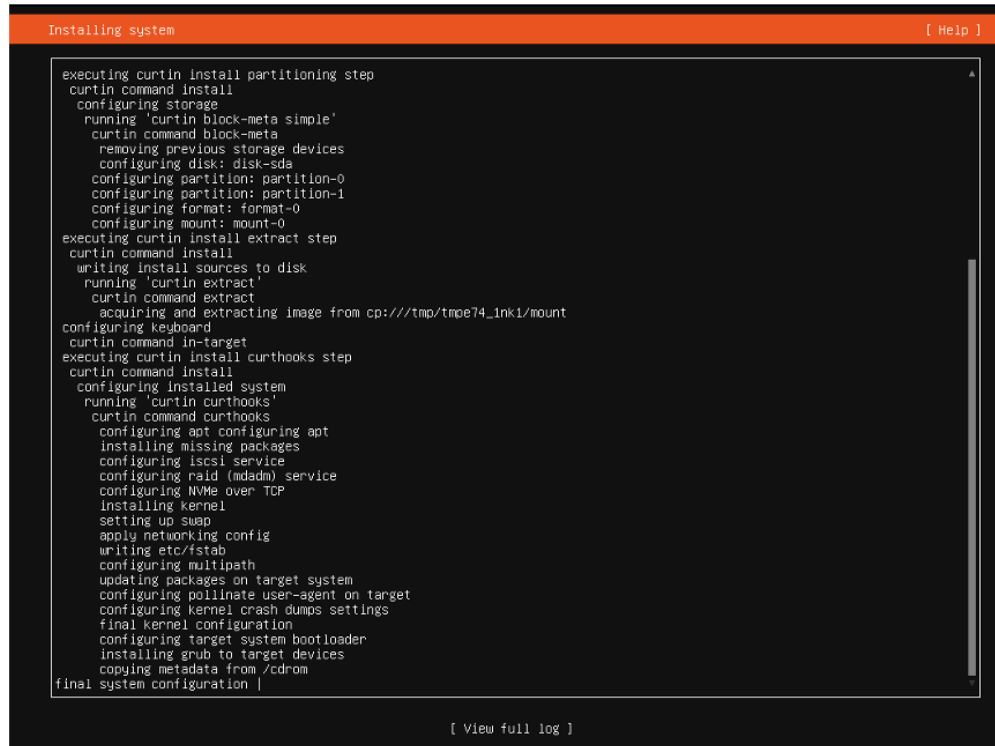
Choose a password:

Confirm your password:

Completed Ubuntu profile setup.

## Task 14: Complete the Software Installation

### Screenshot 1:



```
Installing system [ Help ]

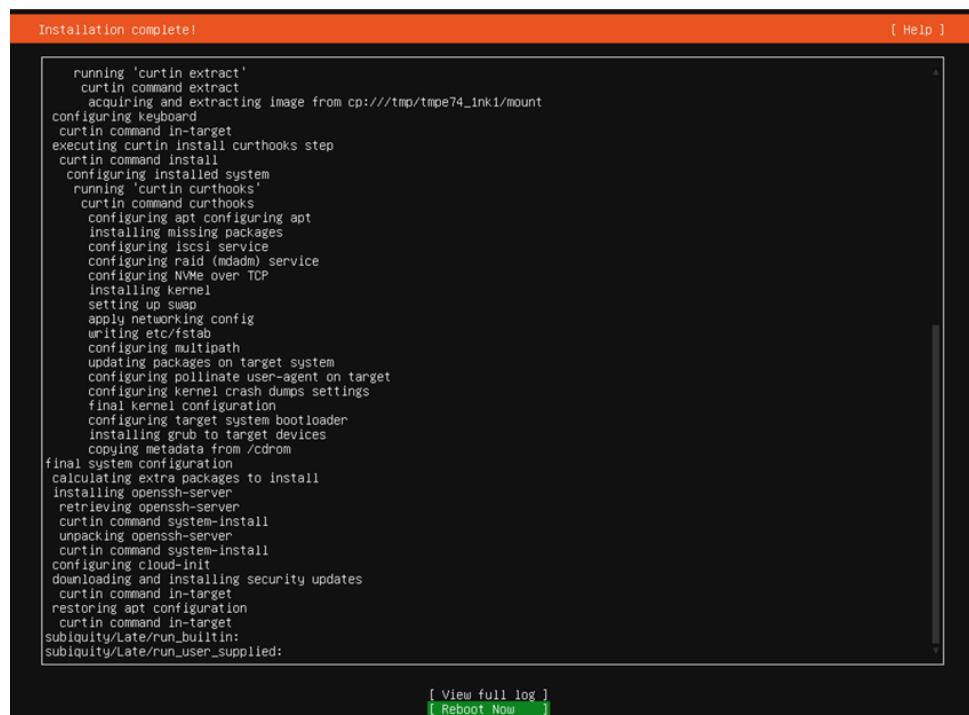
executing curtin install partitioning step
curtin command install
configuring storage
  running 'curtin block-meta simple'
  curtin command block-meta
    removing previous storage devices
    configuring disk: disk-sda
    configuring partition: partition-0
    configuring partition: partition-1
    configuring format: format-0
    configuring mount: mount-0
executing curtin install extract step
curtin command install
  writing install sources to disk
  running 'curtin extract'
  curtin command extract
    acquiring and extracting image from cp:///tmp/tmp74_ink1/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
  configuring installed system
  running 'curtin curthooks'
  curtin command curthooks
    configuring apt configuring apt
    installing missing packages
    configuring iscsi service
    configuring raid (mdadm) service
    configuring NVMe over TCP
    installing kernel
    setting up swap
    apply networking config
    writing etc/fstab
    configuring multipath
    updating packages on target system
    configuring pollinate user-agent on target
    configuring kernel crash dumps settings
    final kernel configuration
    configuring target system bootloader
    installing grub to target devices
    copying metadata from /cdrom
final system configuration ]

[ View full log ]
```

Ubuntu Server software installation progress.

## Task 15: Reboot and Verify the Ubuntu Identity

### Screenshot 1:



```
Installation complete! [ Help ]

running 'curtin extract'
curtin command extract
  acquiring and extracting image from cp:///tmp/tmp74_ink1/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
  configuring installed system
  running 'curtin curthooks'
  curtin command curthooks
    configuring apt configuring apt
    installing missing packages
    configuring iscsi service
    configuring raid (mdadm) service
    configuring NVMe over TCP
    installing kernel
    setting up swap
    apply networking config
    writing etc/fstab
    configuring multipath
    updating packages on target system
    configuring pollinate user-agent on target
    configuring kernel crash dumps settings
    final kernel configuration
    configuring target system bootloader
    installing grub to target devices
    copying metadata from /cdrom
final system configuration
calculating extra packages to install
installing openssh-server
  retrieving openssh-server
  curtin command system-install
unpacking openssh-server
curtin command system-install
configuring cloud-init
downloading and installing security updates
curtin command in-target
  restoring apt configuration
curtin command in-target
subiquity/Late/run_builtin:
subiquity/Late/run_user_supplied:

[ View full log ]
[ Reboot Now ]
```

Ubuntu installation complete screen.

### Screenshot 2:

```
[ 22.241534] cloud-init[1150]: |.#+ .
[ 22.241835] cloud-init[1150]: | .o
[ 22.242109] cloud-init[1150]: +----[SHA256]-----

ubuntu login: ^fatimajaleel
Password:
```

Ubuntu login screen.

### Screenshot 3:

```
ubuntu login: ^fatimajaleel
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Sep 16 07:47:53 PM UTC 2026

System load:  0.04               Processes:           251
Usage of /:   50.5% of 9.75GB    Users logged in:    0
Memory usage: 21%               IPv4 address for ens33: 192.168.67.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

12 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

fatimajaleel@ubuntu:~$ whoami && hostname
fatimajaleel
ubuntu
fatimajaleel@ubuntu:~$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://pk.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:3 http://pk.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://pk.archive.ubuntu.com/ubuntu resolute-backports InRelease
13 packages can be upgraded. Run 'apt list --upgradable' to see them.
fatimajaleel@ubuntu:~$ _
```

Successful Ubuntu terminal login.

### Screenshot 4:

```
fatimajaleel@ubuntu:~$ _
```

Ubuntu terminal showing <github-username>@ubuntu prompt

## Task 16: Find the ubuntu server IP address

### Screenshot 1:

```
fatimajaleel@ubuntu:~$ exit
logout
Connection to localhost closed.
fatimajaleel@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:d3:af:7b brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx00c29d3af7b
    inet 192.168.67.129/24 metric 100 brd 192.168.67.255 scope global dynamic ens33
        valid_lft 1483sec preferred_lft 1483sec
    inet6 fe80::20c:29ff:fed3:af7b/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
fatimajaleel@ubuntu:~$ _
```

### Screenshot 2:

```
inet 192.168.67.129/24 metric 100 brd 192.168.67.255 scope global dynamic ens33
```

## Task 17: Connect to Ubuntu Server from Windows

### Screenshot 1:

```
C:\Users\User>ssh fatimajaleel@192.168.67.129
The authenticity of host '192.168.67.129 (192.168.67.129)' can't be established.
ED25519 key fingerprint is SHA256:gWcFsb+VZEaSaA2kPA2obbs6hApPG+CenARtAJLvHPI.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])?
```

SSH command entered in Windows Command Prompt/PowerShell.

### Screenshot 2:

```
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.67.129' (ED25519) to the list of known hosts.
fatimajaleel@192.168.67.129's password: |
```

Accepted SSH server fingerprint

### Screenshot 3:

```
fatimajaleel@192.168.67.129's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Sep 18 03:46:37 AM UTC 2026

System load:  0.08               Processes:            227
Usage of /:   47.6% of 9.75GB    Users logged in:     0
Memory usage: 28%               IPv4 address for ens33: 192.168.67.129
Swap usage:   0%

 * Canonical Workshop gives developers fast, composable, reproducible, and
   secure developer environments that are perfect for agentic workflows.

   https://ubuntu.com/workshop

Expanded Security Maintenance for Applications is not enabled.

1 update can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

#### Screenshot 4:

```
Last login: Wed Sep 16 20:08:35 2026 from 127.0.0.1
fatimajaleel@ubuntu:~$ |
```

Remote SSH prompt showing <github-username>@ubuntu.